

Runoff

Storm Hydrograph Interpretation

Identifying parts of the hydrograph (Fig. 1):

- **Rising limb:** The initial, relatively steep increase in discharge following rainfall onset.
- **Peak discharge:** The maximum discharge reached during/after the storm.
- **Recession limb:** The gradual decline in discharge as event water inputs diminish and the catchment drains.

Process-based explanation

- **Why is the rising limb steep?** Rapid delivery of water to the channel via quick pathways: direct channel precipitation, overland flow (where it occurs), near-surface lateral flow/“interflow”, and fast subsurface responses (macropores, piston-type displacement). Steepness increases with higher rainfall intensity, high antecedent moisture (reduced storage), steeper slopes, and more impervious surfaces.
- **Why does the hydrograph peak when it does?** Peak occurs when contributing areas and pathways reach maximum effectiveness; typically when rainfall duration $\geq$ time of concentration (the time for flow from the most distant point to reach the outlet). After this, inputs diminish and/or storages begin to dominate outflows, reducing discharge.
- **What dominates the recession limb?** Event water subsides and **subsurface drainage** governs the tail: lateral throughflow and groundwater discharge (baseflow) gradually feed the channel as catchment storages empty.

Influence of antecedent moisture and geology

- **Antecedent soil moisture (ASM):** Wet antecedent conditions reduce infiltration capacity and available storage, yielding **faster, higher peaks** and shorter lags; dry conditions yield more damped responses.
- **Geology/permeability:**
 - **High-permeability (e.g., chalk/limestone, coarse alluvium):** broader peaks, slower rise, **stronger/sustained baseflow** (concave, long recession).
 - **Low-permeability (e.g., clay/shales, bedrock with thin soils):** flashier hydrographs—sharp peaks, quick recessions, weak baseflow.
- **Reasoned inference from Fig. 1:**

- If Fig. 1 shows **flashy peaks and short recession**, infer **low-permeability**/more impervious geology.
- If Fig. 1 shows **damped peaks with strong baseflow and long recession**, infer **high-permeability** geology with substantial subsurface storage.

Runoff Mechanism Comparison

Horton (infiltration-excess overland flow)

- **Dominant control:** Infiltration capacity vs rainfall intensity; overland flow occurs where intensity > infiltration capacity.
- **Sketch features:** Rain > infil. cap. on slopes → thin surface film flows to channel.
- **Likely environments:** Arid/semi-arid storms with intense rainfall; compacted/impervious urban areas; crusted/bare soils.
- **Limitations:** In many humid/forested settings, typical rainfall intensities are **less** than infiltration capacities; continuous sheet flow is uncommon.

Betson (partial contributing areas)

- **Dominant control:** Spatial heterogeneity of infiltration capacity and runoff generation—only certain patches create surface runoff.
- **Sketch features:** Patchy contributing zones connected to drainage network during storms.
- **Likely environments:** Mixed land use/soils; landscapes where only limited areas (e.g., near roads/compacted ground) produce surface runoff.

Hewlett & Hibbert (saturation overland flow)

- **Dominant control:** Water table rising to/near surface; expansion of **saturated zones** in hollows/near channels produces return flow + direct rain-on-saturated ground.
- **Sketch features:** Variable saturated areas growing downslope/near channels during events.
- **Likely environments:** Humid, mid-latitude forested catchments; riparian zones; areas with shallow soil over low-permeability layers.

Synthesis — Variable Source Area (VSA) concept

- Modern view integrates these: **source areas vary in space/time** with antecedent wetness, rainfall, soils, and topography. Surface and subsurface pathways co-exist;

saturated areas near channels expand/contract, explaining much stormflow behavior.

- Supporting definitions of **overland flow** and **interflow** (lateral flow above the water table during storms) align with this mixed-process picture.

Runoff Estimation (Rational Method)

Given

- Catchment area **A** = 80 acres
- Land uses and **C** values:
 - 20% residential single-family: $C = 0.40$
 - 10% streets & driveways: $C = 0.90$
 - 30% lawn (sandy soil): $C = 0.15$
 - 40% woodland: $C = 0.10$
- Design storm intensity **i** = 4.9 mm/hr, and **T_c** = 15 min (so use this i).
- Target units: **m³/s** (SI)

Step 1 – Area-weighted runoff coefficient, C_{basin} :

$$C_{\text{basin}} = 0.20 \times 0.40 + 0.10 \times 0.90 + 0.30 \times 0.15 + 0.40 \times 0.10 \\ = 0.08 + 0.09 + 0.045 + 0.04 = \boxed{0.255}$$

Step 3 – Compute peak discharge (Q):

$$Q = 0.255 \times 4.9 \times 80 = 99.96 \text{ cfs}$$

Step 4 – Sensitivity to urbanisation

Replace the 40% woodland ($C = 0.10$) with parking lots ($C \approx 0.90$):

$$C'_{\text{basin}} = 0.20(0.40) + 0.10(0.90) + 0.30(0.15) + 0.40(0.90) = 0.08 + 0.09 + 0.045 + 0.36 = \boxed{0.575}$$

$$Q' = 0.575 \times 4.9 \times 80 = \boxed{225.39 \text{ cfs}}$$

Increase factor: $(0.113/0.0502 \approx \mathbf{2.25})$.

Groundwater and Subsurface Contributions

A. *Piston flow vs throughflow*

- **Throughflow:** Lateral movement of water through the **unsaturated** soil matrix on slopes, contributing to streams during/after storms.
- **Piston flow (conceptual):** New rainfall infiltrates at the top and **displaces older pore water** downslope/downslope interfaces (e.g., along a low-permeability horizon), pushing it rapidly toward streams—akin to a pressure pulse.
- Result: Event hydrographs can contain a large fraction of **pre-event (“old”) water** mobilised from storage rather than newly infiltrated rain.

B. *Capillary fringe/“groundwater ridge” contribution*

- During recharge, rising heads near channels create a local **groundwater mound** (ridge) adjacent to streams, steepening hydraulic gradients toward the channel.
- This accelerates **rapid groundwater discharge** to the stream, boosting stormflow beyond what overland flow alone can explain.

C. *Why is “old water” observed after storms? Implications*

- Tracer studies show much stormflow is stored water quickly **mobilised by pressure and gradient changes** (piston-like displacement, macropore bypass, rising water tables), not solely new rain.
- **Implication:** Subsurface pathways and dynamic saturated areas are central to storm response; the hydrograph recession reflects **drainage of subsurface storages** (throughflow and groundwater).
- Overland flow and interflow are part of **direct runoff**, but the significant role of subsurface contributions explains sustained recessions and the presence of old water in early stormflow.